



Happy Birthday foss4g.be !

State of



*Or: a 20
minute pitch
about why
GRASS is
(often) the
best solution
;-)*

Moritz Lennert

osgeo.be, former GRASS developer and PSC member

inspired/stolen from

Alen Mangafić, Vaclav (Vashek) Petras and Corey White

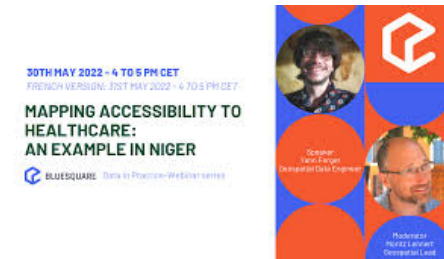
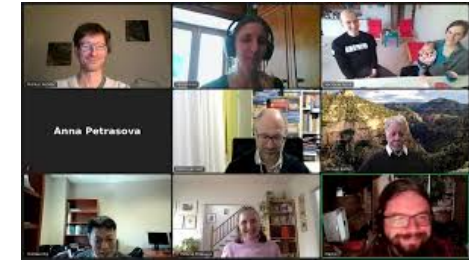
Geodetic Institute of Slovenia, NCSU GeoForAll

FOOS4G Belgium – September 25, 2025 – Brussels



Why me, why GRASS ?

- Geographer (ULB & Bluesquare)
- GRASS developer for almost two decades
- University researcher and teacher of GIS with GRASS
- Now: GIS-retired, but still proud founding member of osgeo.be !



- **GRASS is like good Belgian beer:** first contact might not always be the easiest because it's so strong, but once you're used to it, you're hooked !
- Why ? High diversity of functionalities, very reliable geospatial processing engine, HPC computing, long-term stability AND innovation



Focus on some important areas of development

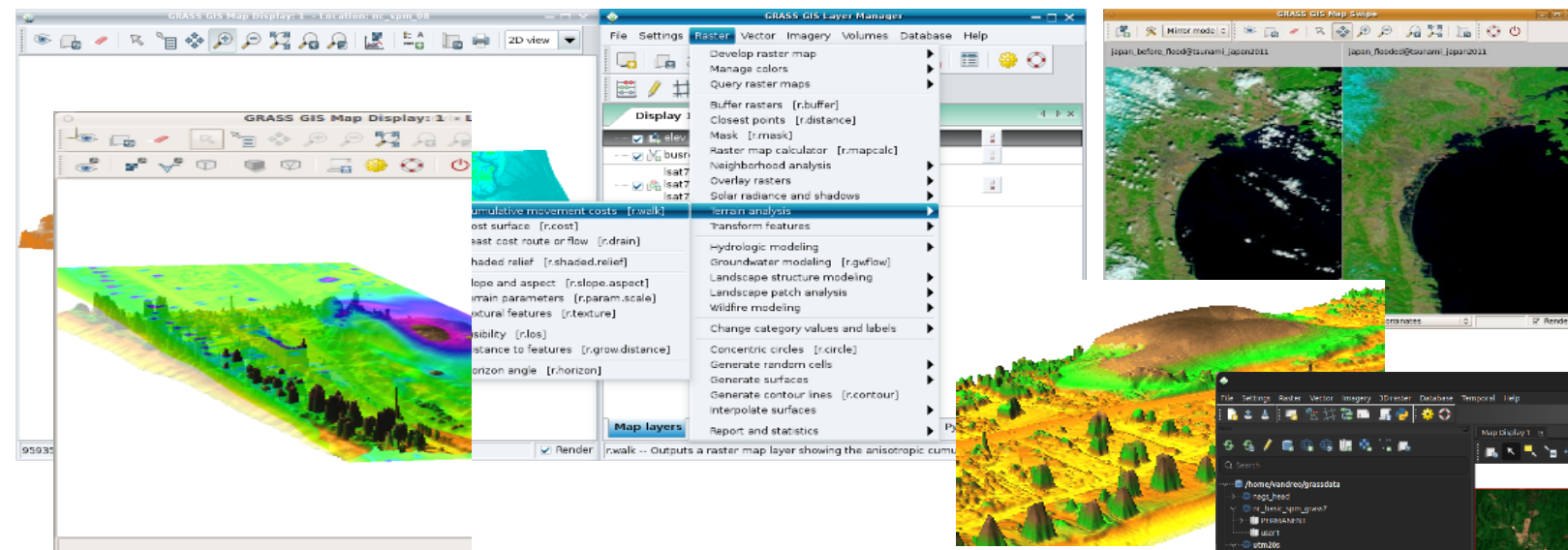
- Improving the user experience
- A renewed community of users and developers
- Continuous innovation
- GRASS as a HPC geospatial processing engine

Improving the user experience

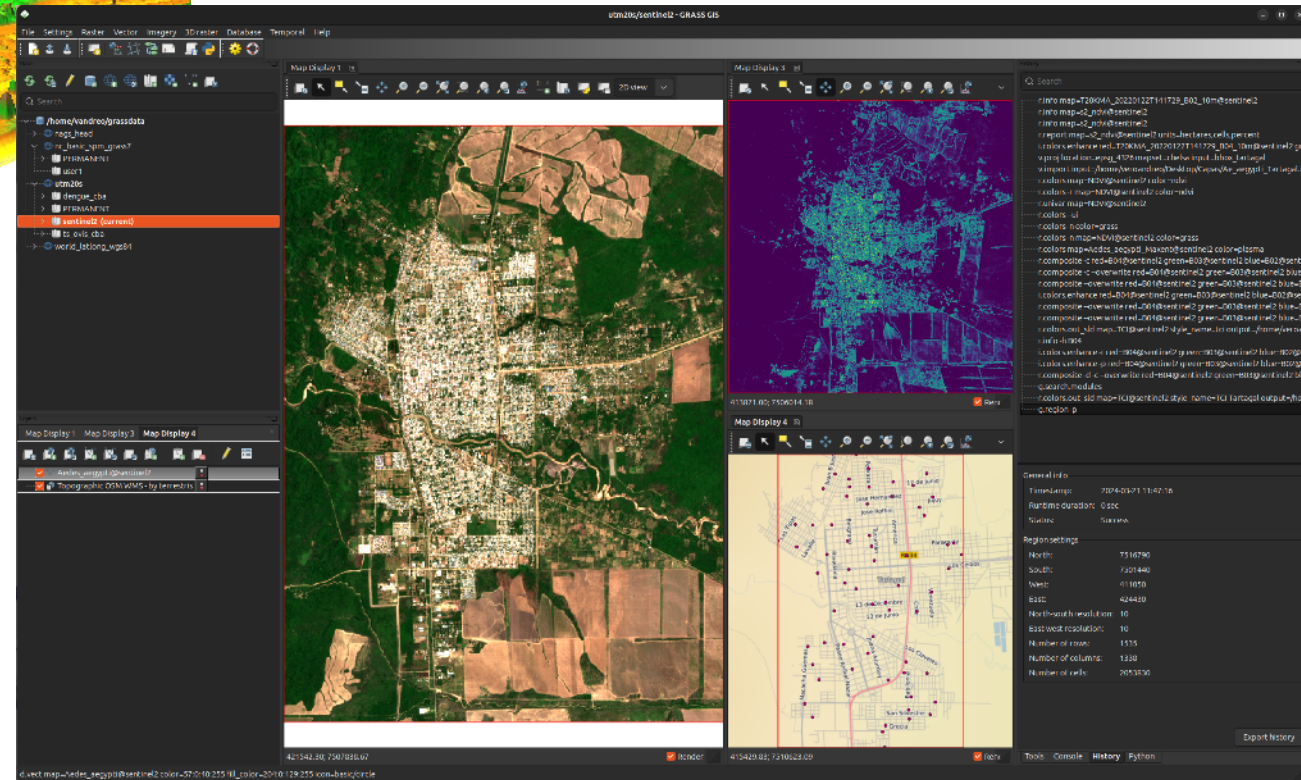
- Completely reworked GUI
 - Single-window
 - More tools
- Easier startup
- Location → Project

Single window GUI

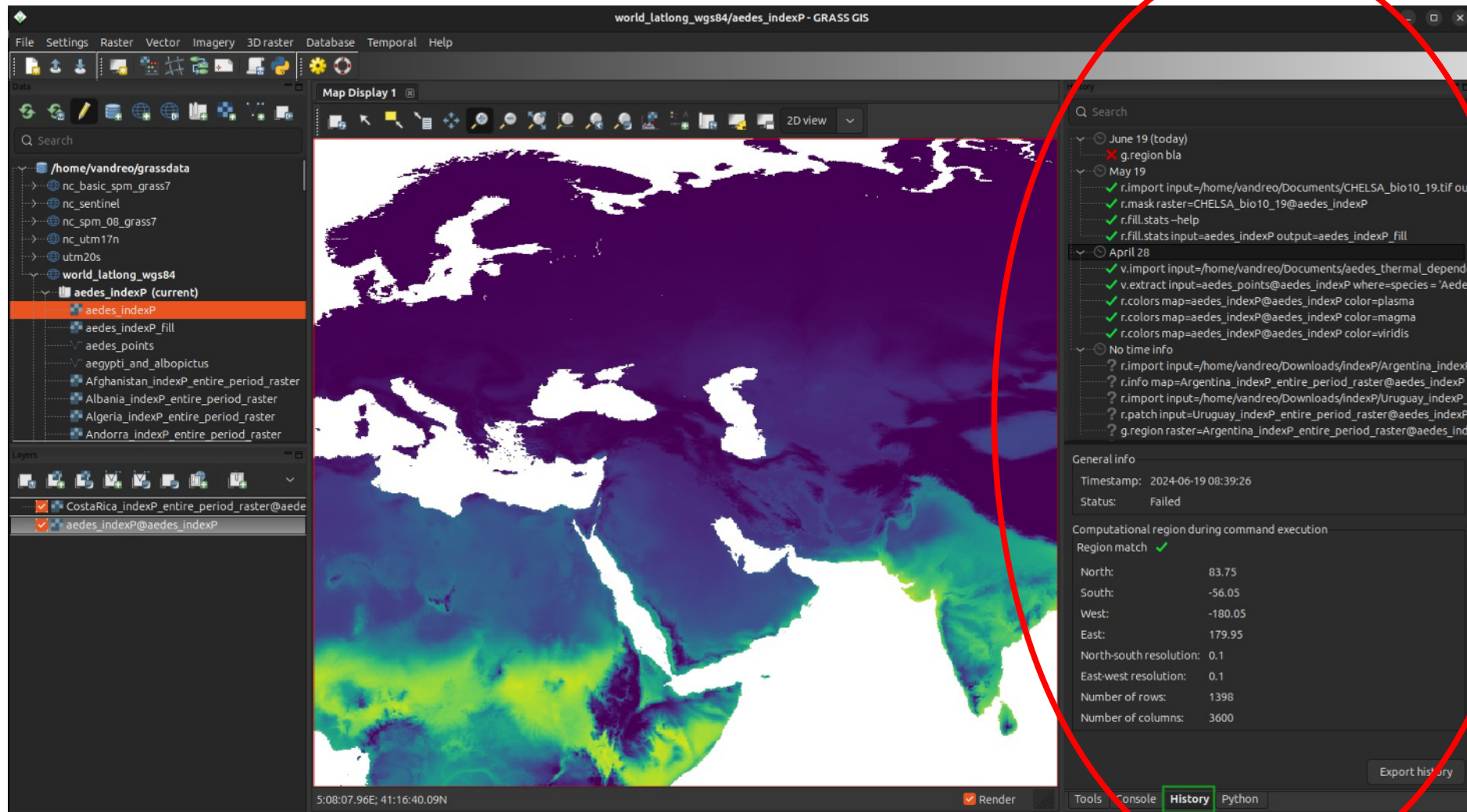
GRASS presented at beOpenGis in 2014 :



GRASS now :

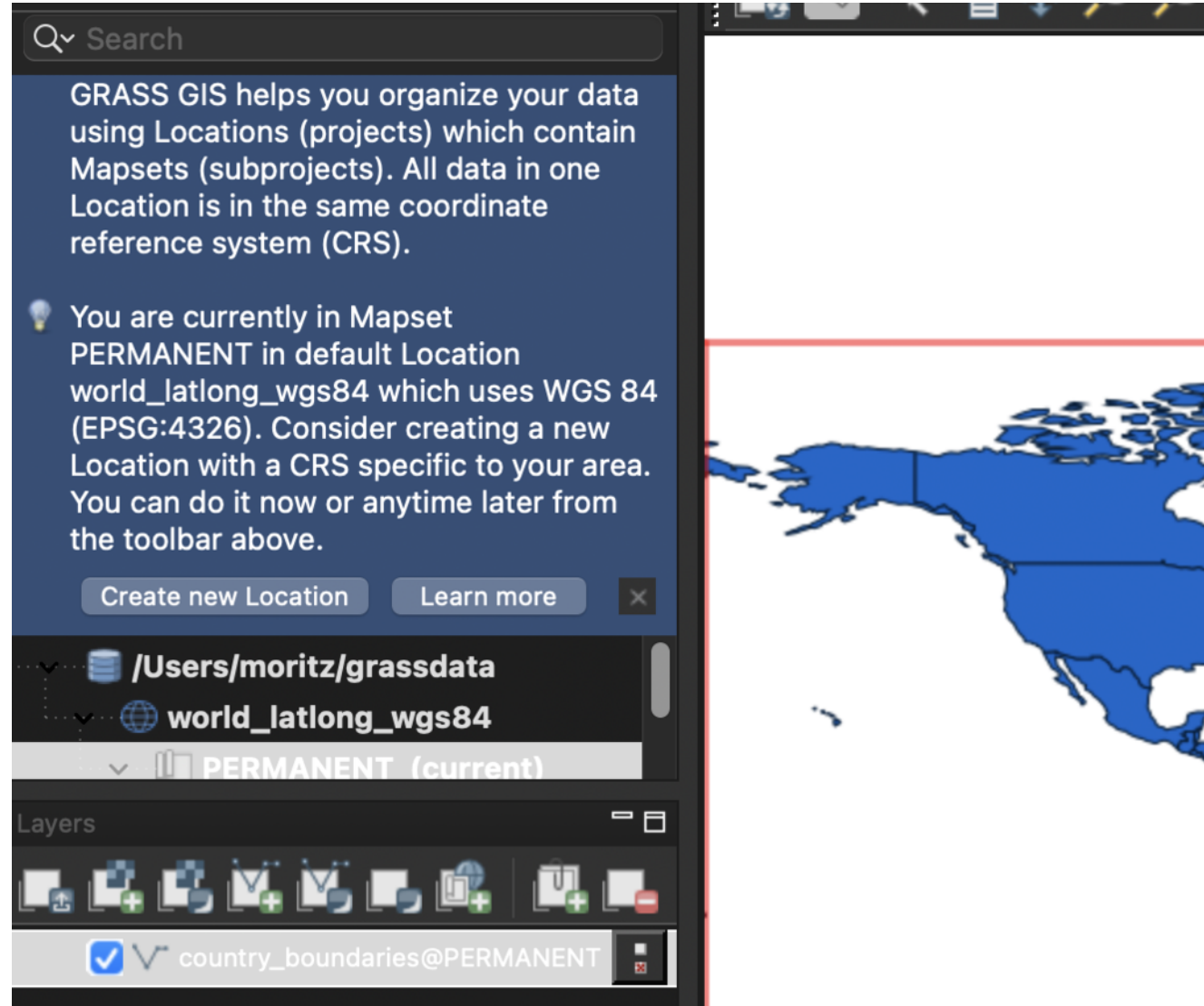


GUI: new History Browser panel



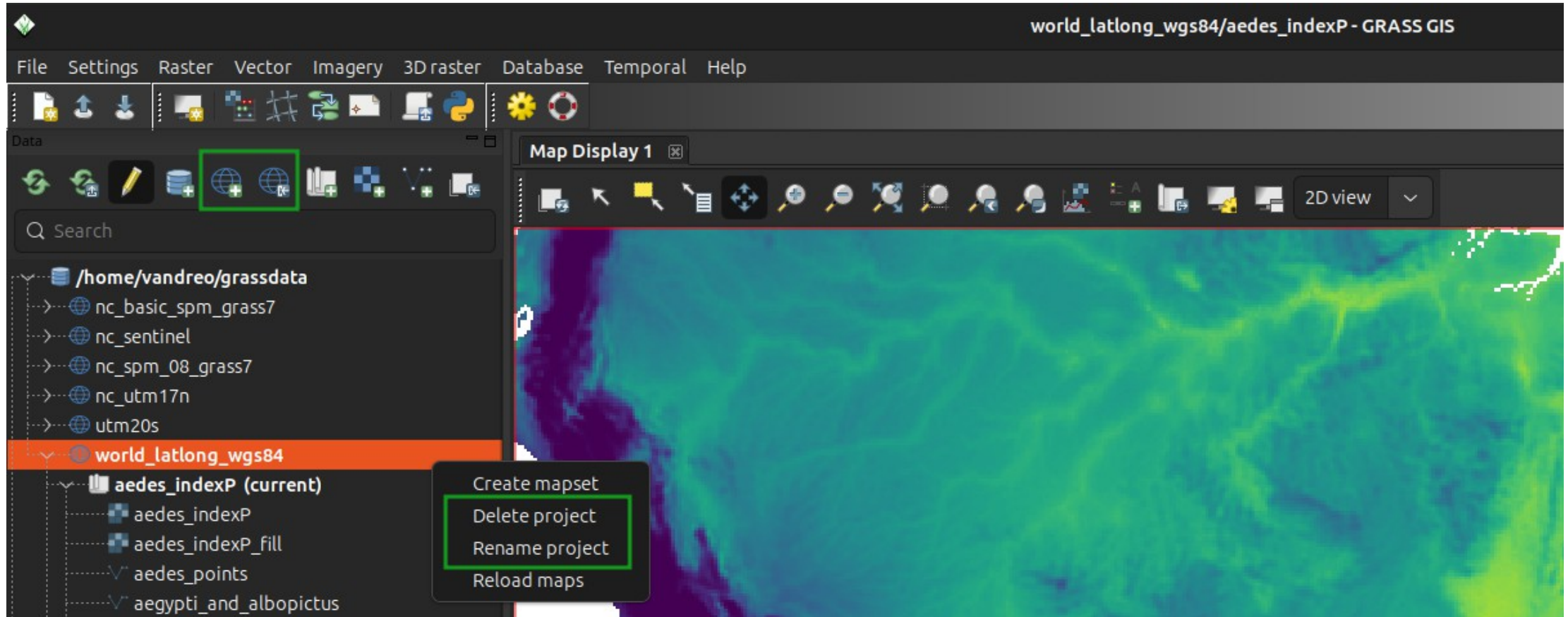
by Linda Karlovská with a GRASS Student Grant

Easier startup



Shaking up the foundations ?

Locations are now Projects



GRASS is for old people ?

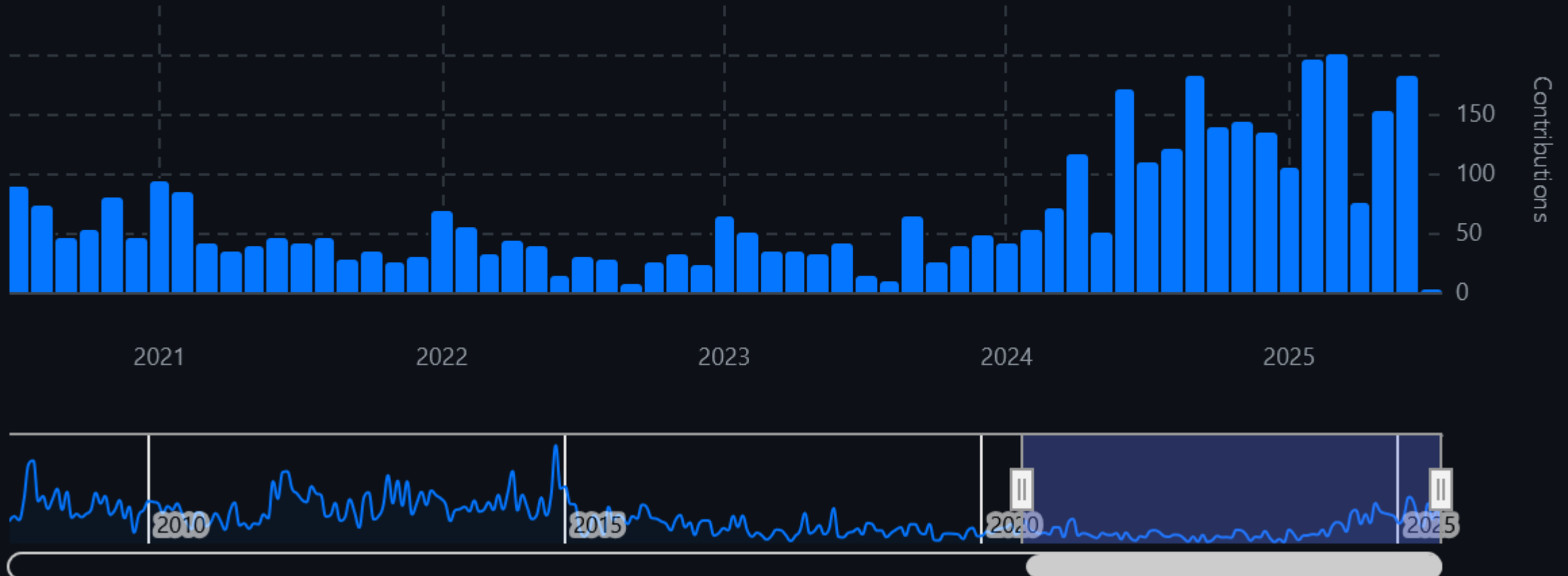
Come and see the community !

- Renewal of developer community
- Large US grant
- Significant reworking of internals
- Constant development of new addons

Community

Commits over time

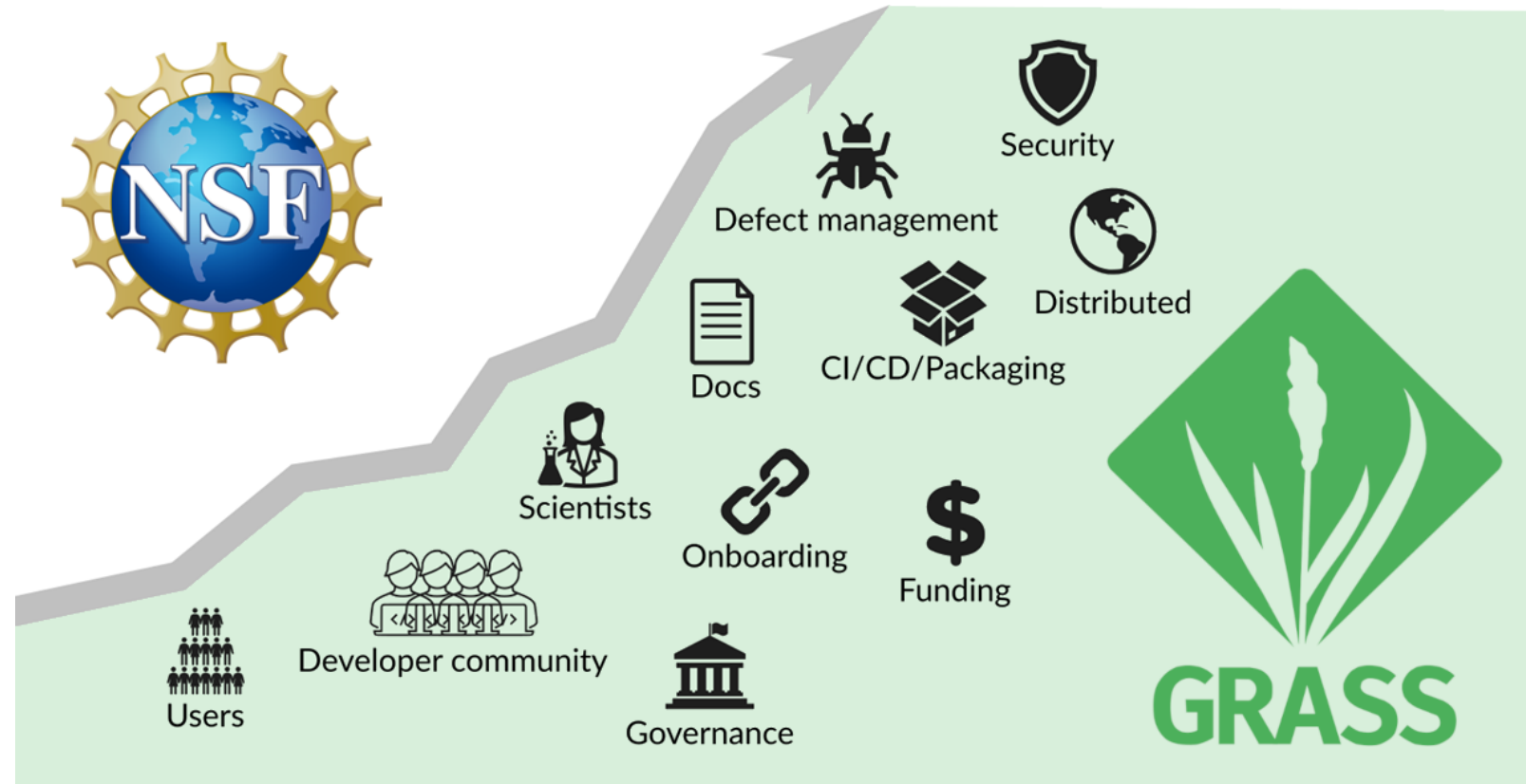
Weekly from 4 Jul 2020 to 6 Jul 2025



NSF Grant Lead by NC State

- NSF grant awarded to NC State, ASU, NMSU, Yale
 - To enhance infrastructure
 - To revise contributing guidelines
 - To support community building

The NSF program is not funding new features, bug fixes, or ongoing maintenance.



GRASS Developer Summit 2025

Never too young to
start developing for
GRASS !



grass.osgeo.org > Latest news > Report from the GRASS Community Meeting
Funding: US NSF (Award 2303651), Bohannon Huston, Inc., and the Center for Geospatial Analytics at NC State

Integrating Students

Grants Program: Student grants to contribute to GRASS (continuous call every year)



Long history of Google Summer of Code projects

Google Summer Of Code 2024

Monday, May 27, 2024

Three GSoC students will be contributing to GRASS GIS this summer!

- GRASS GIS Interactive Scatter Plot tool
- Temporal GIS Algebra for raster and vector data in GRASS
- 2012
 - Python high level map interaction for GRASS GIS
 - GRASS GIS WxGui front end for vector analysis modules
 - Image Segmentation in GRASS GIS
- 2011
 - Completion of wxGUI Nviz extension for 3D data visualization in GRASS GIS
 - Graphical User Interface for the hydrological tools r.stream* in GRASS GIS
 - GRASS wxGUI WMS service rendering
 - r.in.modis for GRASS GIS
- 2010
 - wxNviz development for enhanced 3/4D visualization and analysis
 - GPU accelerated imagery resampling and reprojection
- 2009
 - GRASS - Network Analysis
 - v.autokrige: Python porting and wxPython GUI addition
 - OssimPlanet integration in Grass and Qgis
- 2008
 - Reimplement And Add Features to Buffer Generation Module in GRASS
 - 2.5/3D visualization tool for wxPython GRASS GUI
 - Reimplementation of v.voronoi and v.delaunay modules in the Vector ...
- 2007
 - GRASS Modules for line generalization and smoothing
 - Shortest path in free (vector) space avoiding obstacles module in GRASS

GSoC-2025 Introduction: N

Initiatives Google Summer of Code grass



Nishant

Hi Everyone,

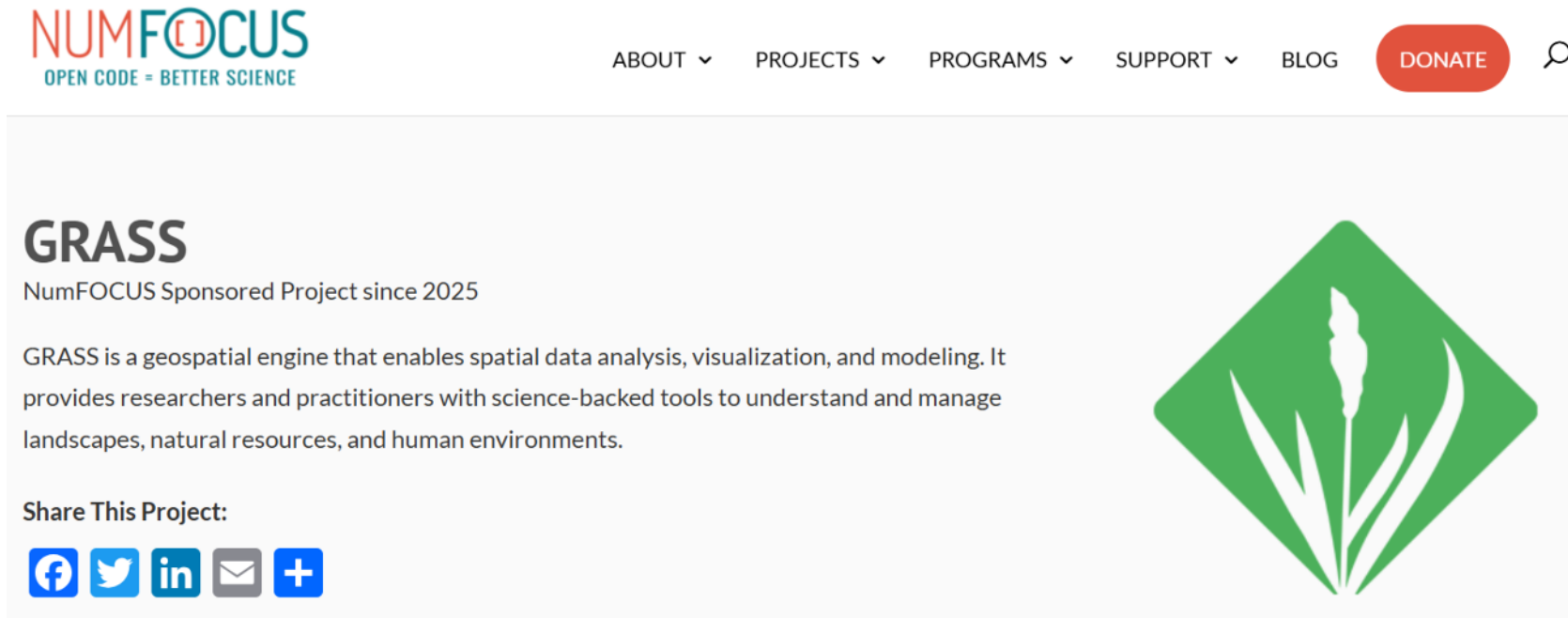
I'm Nishant Bansal, an undergraduate student from Varanasi. I'm honored to have been selected for the GRASS GIS project under the mentors.

This summer, I'll be enhancing GRASS GIS tools. The project plan includes:

ation

Financial Contributions

- Contributions through NumFOCUS partially support face-to-face meetings and student grants.
- We would like to finance ongoing maintenance.



The screenshot shows the NumFOCUS website header with the logo "NUMFOCUS OPEN CODE = BETTER SCIENCE" and navigation links: ABOUT, PROJECTS, PROGRAMS, SUPPORT, and BLOG. A red "DONATE" button is also visible. The main content area features the "GRASS" project title, a subtitle "NumFOCUS Sponsored Project since 2025", and a description: "GRASS is a geospatial engine that enables spatial data analysis, visualization, and modeling. It provides researchers and practitioners with science-backed tools to understand and manage landscapes, natural resources, and human environments." A green diamond-shaped logo with a white plant-like design is shown to the right. Below the description, there is a "Share This Project:" section with social media icons for Facebook, Twitter, LinkedIn, Email, and a generic share icon.



<https://numfocus.org/project/grass>

More and more commercial Support



Margosa Environmental Solutions Ltd

Margosa develops pioneering geodata solutions using advanced open source technologies and machine learning frameworks, which enables the integration of massive environmental and GIS datasets. We offer cost-effective, scalable, and secure global data analytics platforms for government, multilateral, corporate, educational, and not-for-profit institutions. Our mission is to transform complex natural resource information into practical knowledge for decision-makers and stakeholders alike.

[VISIT OUR WEBSITE](#)



Kan Territory & IT

Kan Territory & IT develops geographic solutions based on the integration of different open source software packages including GRASS and orchestration tools to facilitate the management and analysis of geospatial data for decision making. This powerful combination has completely transformed our clients' geoprocessing workflows, delivering efficiency and accuracy.

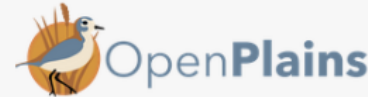
[VISIT OUR WEBSITE](#)



mundialis GmbH & Co. KG

At mundialis we work with free geodata and free software in the field of remote sensing and GIS analysis. We generate spatial information and develop free and open source software. Both form the core of customised GIS solutions that we implement in a customer-oriented manner. We work scientifically based with regional and international project partners. In addition to our technical work, we see ourselves as an environmentally conscious and sustainable company that makes a positive contribution to society and the environment.

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OpenPlains Inc.

OpenPlains Inc. provides the best decision making tools through the development of intuitive interactive geospatial modeling applications. Let us help you put your GRASS models into production.

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OpenGeoLabs Ltd.

OpenGeoLabs Ltd. is a software company based in the Czech Republic. We focus on developing, consulting, lecturing, and mentoring open-source geospatial software. We are building a large team of external consultants, developers, and experts so that we can provide the highest quality services. Our team members contribute to software such as GRASS, QGIS, GDAL, PyWPS, OWSLib, and PostGIS, among others. We are active members of the global open-source community through OSGeo, as well as on a local level.

[VISIT OUR WEBSITE](#)

Center for
Geospatial Analytics

NC STATE UNIVERSITY

Center for Geospatial Analytics at NC State University

At the Center for Geospatial Analytics, we develop and improve GRASS open source geospatial software. Our [Service Center](#) builds solutions leveraging GRASS GIS and provides training.

[VISIT OUR WEBSITE](#)

Continuous innovation

- Constant flow of new add-ons developed
 - Needs
 - Cutting-edge research
- Complete renewal of documentation logic

Google Scholar



GRASS

OSGeo

Verified email at osgeo.org - [Homepage](#)

[GIS](#) [Geographic Information Sy...](#) [Remote Sensing](#) [Geoinformatics](#) [Open Source](#)



TITLE	CITED BY	YEAR
Geospatial physiologically based demographic modeling and analysis of thirteen invasive species in Africa: why the biology matters AP Gutierrez, L Ponti, M Neteler, F Stocchino, JR Cure, PE Kenmore, ... Environmental Sciences Europe 37 (1), 1-25	3	2025
Multiple factors controlling polar willow growth in the high Arctic (Svalbard): implications for future prediction of tundra productivity M Phulara, M Opala-Owczarek, W Szymański, J Ślopek, M Węgrzyn, ... Global Ecology and Conservation, e03801		2025
Relationship Between Aggregation Index and Change in the Values of Some Landscape Metrics as a Function of Cell Neighborhood Choice P Zatelli, C Tattoni, M Ciolli ISPRS International Journal of Geo-Information 14 (8), 304		2025
Combination of statistical and conceptual approaches for debris-flow susceptibility modelling at a regional scale, British Columbia, Canada T Bornaetxea, A Blais-Stevens, B Miller, I Marchesini Catena 256, 109044	1	2025
An integrated remote sensing lineament investigation of the Manicouagan region, Canada: Implications for planetary cratering processes MA Fobert, JG Spray, V Singhroy Journal of Geophysical Research: Planets 130 (8), e2025JE008962		2025
A Cloud-Based Geoprocessing Platform Based on Extended OGC API Processes Standard		2025

Some examples of recent new addons

- **r.slopeunits**: toolset that creates, cleans and calculate metrics for slope units
- **r.hydrobasin**: OpenMP parallel watershed delineation based on Cho (2025)
- **r.flowaccumulation**: Flow accumulation using Memory-Efficient Flow Accumulation parallel algorithm by Cho (2023).
- **r.in.vect**: new addon to import a vector map as raster layer
- **r.buildvrt.gdal**: Create GDAL VRTs for GDAL linked raster maps
- **i.eodag**: Downloads imagery datasets from various providers through the EODAG API.
- **r.fusion**: image fusion, generalized pan-sharpening.(funded by mundialis GmbH & Co. KG)
- **r.windfetch**: addon for computing wind fetch.
- **r.maxent.train** & **r.maxent.predict**: train, and predict a Maxent model to create a suitability distribution layer

400+ processing tools, 400+ more in the form of addons !

New documentation

**GRASS 8.5.0dev Documentation**

Search

OSGeo/grass
8.4.1 1k 368

Getting started Interfaces Key concepts Tools Development

Getting started
Getting started
Tutorials

Getting started

What Is GRASS?

GRASS is a geospatial processing engine for advance analysis and visualization of geospatial data. GRASS is a free and open source software, released under the GNU GPL.

[Download GRASS](#)

[Grow GRASS](#)

Tools

GRASS provides a wide range of tools for geospatial processing, modeling, analysis, and visualization, with detailed documentation to guide users through processing raster, vector, 3D raster, and temporal data.

Explore tools →

Interfaces

GRASS provides a number of interfaces for interacting with the tools and developing computational workflows. The most common interfaces are:

- Command line, also know as terminal or shell
- Python

Table of contents
What is GRASS?
Tools
Interfaces
Key concepts
Development

New Tutorial Site

grass-tutorials.osgeo.org



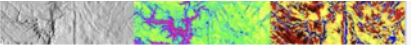
Earthworks: Terrain Synthesis

EARTHWORKS TERRAIN RASTER

INTERMEDIATE

Learn how to synthesize terrain with `r.earthworks`.

BRENDAN HARMON JUN 30, 2025



Visualizing and Modeling Terrain from DEMs in GRASS

BEGINNER INTERMEDIATE RASTER

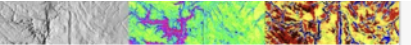
DEM VISUALIZATION TERRAIN SLOPE

ASPECT LANDFORMS HYDROLOGY

STREAMS

This tutorial guides a user through multiple ways of analyzing and modeling terrain from DEM raster maps.

EUNICE VILLASEÑOR IRIBE JUN 23, 2025
AND MICHAEL BARTON



Visualização e modelagem de terrenos a partir de MDE no GRASS

BEGINNER INTERMEDIATE RASTER

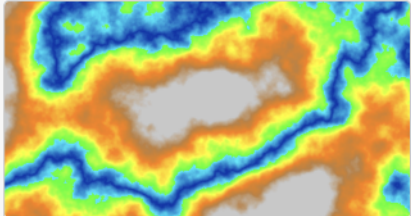
DEM VISUALIZATION TERRAIN SLOPE

ASPECT LANDFORMS HYDROLOGY

STREAMS PORTUGUÊS

Este tutorial orienta o usuário por meio de várias maneiras de analisar e modelar terrenos a partir de mapas raster MDE.

EUNICE VILLASEÑOR IRIBE, JUN
MICHAEL BARTON, & LETICIA 23,
CORREA 2025

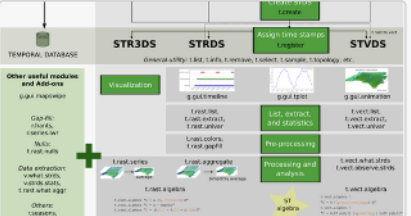


Basics of map algebra

MAP ALGEBRA RASTER TERRAIN

BEGINNER

Learn the basics of map algebra in GRASS including arithmetic, functions, and logic. Use map algebra to model a landscape.

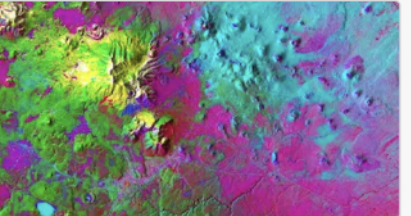


Deep dive into time series analysis with GRASS

TIME SERIES RASTER INTERMEDIATE

ADVANCED PYTHON

A collection of tutorials demonstrating how to handle time series data in GRASS, from basic concepts to



Introduction to Remote Sensing with GRASS

BEGINNER GUI IMAGERY

REMOTE SENSING VISUALIZATION

Analysis and visualization of multi-band satellite imagery using image fusion, environmental indexes, and

lead by the NC State Center for Geospatial Analytics

Surfing the cloud: GRASS as a HPC geospatial processing engine

- Stability
- Efficiency
- Programmability

Stability (or how to keep your mental health)

- Invest your time into technologies which stay and don't disappear in couple years.
- Bash scripting, Python, GRASS.
- Longevity, open-source, platform-independent.
- Things develop, versions update, features are added...
...but general ideas stay the same.

Code from 2002 running in 2025?

version 5.0 code works as is in 8.4.1:

```
r.mapcalc depr.bin="if((elev - fill)< 0., 1, 0)"
```

Efficiency

- Parallization
- C-level within-module (thank you GSoC !)
- Easy spatial parallization between modules and module runs



=WALOUS

Creating Wallonia's new
very high resolution land cover maps:
combining GRASS GIS OBIA
and OTB pixel-based results

FOSS4G 2019

M. Lennert, T. Grippa, J. Radoux, C. Bassine, B. Beaumont, P. Defourny, E. Wolff

The authors would like to acknowledge the Walloon Government for the funding of the project WALOUS.

25cm orthophotos (RGB + NIR)
of all of Wallonia – 2.5TB of
data (presented at foss4g.be
2019)

Programmability: easier integration of GRASS into scripts

grass.script Python package:

Greatly simplified the creation of new projects in Python without a running GRASS session

```
import grass.script as gs
gs.create_project("cordoba_utm21s", epsg="32721")
gs.setup.init("cordoba_utm21s")
```

by Vaclav Petras, funded by NSF Award #2322073, granted to Natrx, Inc.

CLI: temporary project with --exec

Or directly work in temporary data structures:

```
grass --tmp-project EPSG:3358 --exec \
  r.viewshed input=elevation \
  output=viewshed coordinates=642964,222890
```

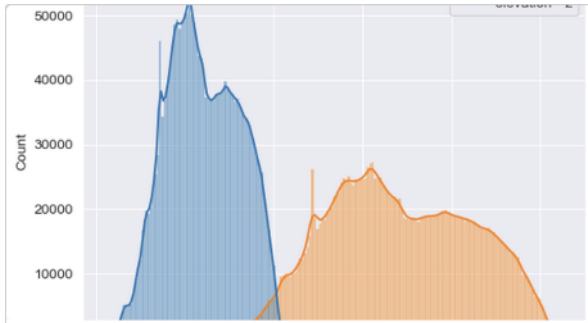
CLI: json-format module output

JSON output format support (format="json") in multiple tools

```
import json
import grass.script as gs

data = gs.parse_command("v.db.select",
                        map="roadsmajor", format="json")
for row in data["records"]:
    print(row["ROAD_NAME"])
```

Programmability: APIs and development environments



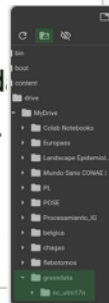
Get started with GRASS & Python in Jupyter Notebooks

PYTHON BEGINNER

Learn how to analyze geospatial data with Python in Jupyter Notebooks.

VERONICA ANDREO

MAR 25,



Get started with GRASS in Google Colab

PYTHON GOOGLE COLAB BEGINNER

Learn how to analyze geospatial data with Python in Google Colab.

VERONICA ANDREO

APR 12, 2024

actinia: cloud based geoprocessing engine

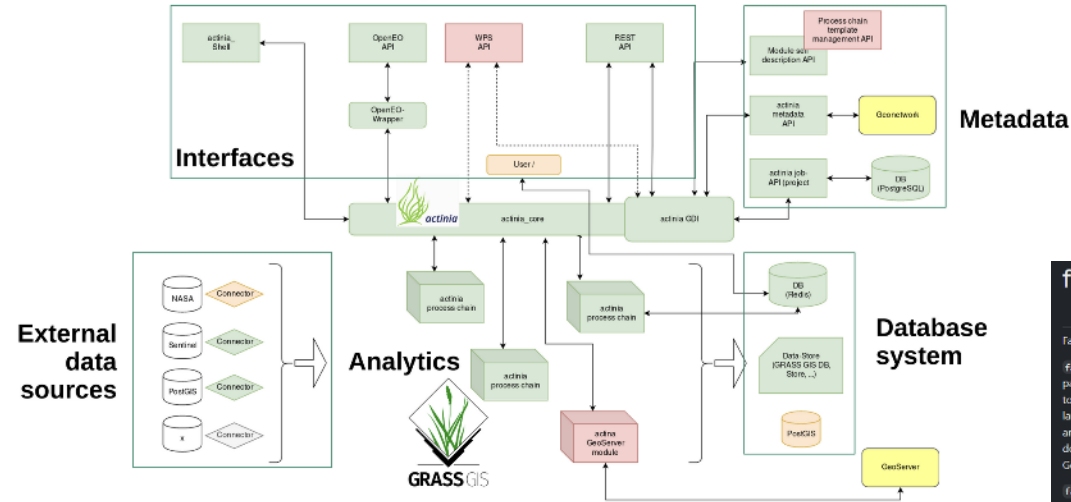
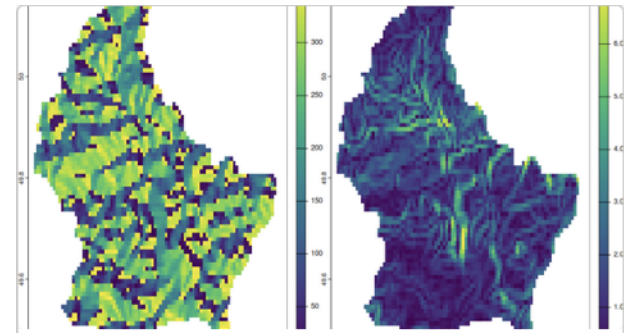


Image source: neteler.gitlab.io/actinia-introduction/



Get started with GRASS & R: the rgrass package

R RGRASS INTERMEDIATE

Learn how to use GRASS and R to analyze geospatial data.

fasterRaster

faster raster processing in R using GRASS

fasterRaster is an R package designed specifically to handle large-in-memory/ large on disk spatial rasters and vectors. fasterRaster does this using Open Source Geospatial's GRASS.

fasterRaster was created with five design principles:

- **Value added:** fasterRaster complements terra and sf, and is highly dependent on them! It is useful for analyzing large-in-memory/large on disk rasters and vectors that those packages struggle to handle. For medium- and small-size objects, terra and sf will almost always be faster.

- **Familiarity:** If you know how to use terra, you basically know how to use fasterRaster! That's because most of the functions have the same name and almost the same arguments as terra functions.



Links
[View on CRAN](#)
[Browse source code](#)
[Report a bug](#)
License
GPL (>= 3)
Community
[Code of conduct](#)
Citation
[Citing fasterRaster](#)
Developers
Adam B. Smith
Maintainer, author
Dev status
CRAN 5.5.19
R CMD Check passing
License GitHub

9, 2024

Programmability

- Upcoming: integrating different workflow tools coherently in the GUI
- ⇒ different easy entry points into programming with GRASS

lindakarlovs

Following recent discussions, I have a broader idea that could be implemented in the GRASS GUI.

Workflow Trio

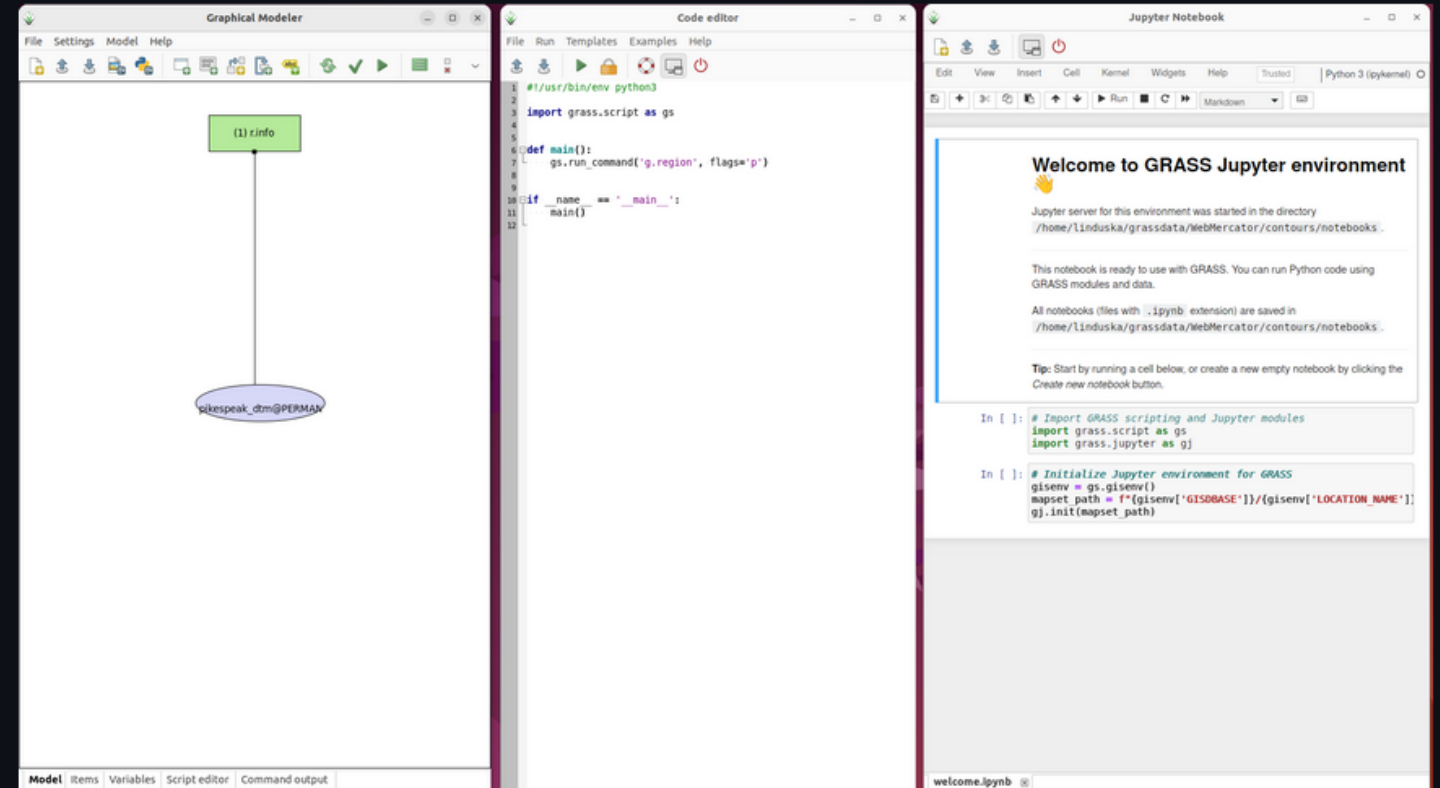
While presenting the idea, I realized it's not "just" a GUI.

- Graphical Modeler
- Simple Python Editor (Code Editor)

Workflow Trio in GRASS

- Graphical Modeler (✅ Available)
- Simple Python Editor (✅ Available)
- Jupyter Notebook (⚠️ Coming soon – integration in progress)

GUI





Current stable: 8.4.1

Version 8.5 coming out soon



As good as Belgian beers !



Try it out !

grass.osgeo.org